Engineering Chemistry (AHT-002) - Unit 2

Unit 2 - Thermodynamics & Electrochemistry - notes in easy English, matched to the UTU syllabus.

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What this unit covers (syllabus): first and second laws of thermodynamics; energy, entropy and free energy; EMF and cell potential; the Nernst equation and applications; acids and bases; Ellingham diagrams in metallurgy; Hess's law.

Key ideas: First law: energy is conserved - $\Delta U = q + w$. **Second law:** entropy of the universe increases for a spontaneous process. **Gibbs free energy** $\Delta G = \Delta H - T\Delta S$ decides spontaneity: $\Delta G < 0$ spontaneous. **Electrochemistry:** a galvanic cell converts chemical energy to electricity; **EMF** = E(cathode) - E(anode), and $\Delta G = -nFE$ connects the two halves of the unit. The **Nernst equation** $E = E° - (0.0591/n) \log Q$ (at 25°C) corrects for concentration - the standard numerical. **Hess's law:** total enthalpy change is the same whatever the path - add/subtract given equations to reach the target. **Ellingham diagram** plots $\Delta G°$ vs T for oxide formation; the line lower at a given temperature reduces the oxide above it - this explains choice of reducing agents in metallurgy.

Exam pointers: one Nernst-equation numerical (calculate EMF at given concentrations), $\Delta G = \Delta H - T\Delta S$ spontaneity checks, and a Hess's law calculation are the regulars. "Explain the Ellingham diagram and its use in metallurgy" is a frequent long answer.

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